



FPGA implementation of modified error normalized LMS adaptive filter for ECG noise removal

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Received: 20 October 2017 / Revised: 15 December 2017 / Accepted: 20 December 2017 / Published online: 22 January 2018
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Abstract

High frequency noise and channel noise are dominant in wireless ECG monitoring systems which can be modeled as white Gaussian noise. Least mean square (LMS) algorithm based adaptive filters are the preferred choice for white Gaussian noise removal, because they require fewer computations and less amount of power consumption. Though LMS algorithm is simple to implement in real time systems, it is necessary to modify the LMS algorithm to reduce the mean square error for improved filtering performance. In this paper, a delayed error normalized LMS (DENLMS) adaptive filter is studied with pipelined architecture to remove the white Gaussian noise from ECG signal. The pipelined VLSI architecture is utilized to boost the operational speed of adaptive filter by reducing the critical path using delay elements. The performance of pipelined DENLMS algorithm is compared with ENLMS and DNLMS algorithms. The pipelined DENLMS filter increases the speed of operation and reduces power consumption at the cost of increase in area due to the presence of latches. Virtex 5 FPGA XC5LVX330 Field programmable gate array has been utilized as target chip to determine the speed, logic utilization and power consumption.

Keywords Adaptive algorithms · Gaussian noise · Field programmable gate arrays · Pipeline processing · Least mean square methods

1 Introduction

Electrocardiogram (ECG) signal preprocessing is essential in biomedical applications to remove unwanted noisy components and to enhance the desired feature extraction. High-quality ECG data is required for improved clinical diagnosis and better treatment of cardiac diseases. However, in real situations, the recorded ECG signals are affected by baseline wander, motion artifacts and high frequency noise. High-frequency noises are occurred in ECG signals recordings due to power line interferences, electromyogram (EMG) noise and electrodes. Baseline wander arises due to the fact that there is a possible movement of either patients or recording cables. Moreover, ECG signals are affected by

white noise due to poor channel conditions in biotelemetry applications [1]. In view of these interruptions and noises during acquisition, ECG preprocessing techniques are focused mainly towards noise removal. Several ECG enhancement methodologies have been discussed in the existing works that are mainly concentrated on suppression of noises such as baseline wander, high frequency noise and white noise using various signal processing tools and algorithms. Some of the techniques are advanced averaging, wavelet transform, non-linear filter banks and adaptive filters [2,3].

In telemedicine applications, transmission of ECG signals over a wireless channel is often affected by noises due to improper channel. The noises are normally modeled as white Gaussian noise [4]. In some applications, white Gaussian noise is considered as a general frequency noise component which is intentionally included to the uncontaminated ECG signals. Chang and Liu [5] proposed a white Gaussian noise removal technique using Wiener filtering approach [5]. Wiener filter is an optimal filter and is mainly used in place of traditional low pass filters where the input signal power spectrum and noise spectrum cannot be separated [6]. Extended Kalman filter (EKF) is an alternative filtering approach to

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conventional Kalman filter that is based on nonlinear signal processing of noisy ECG data [7].

Adaptive filters are used to overcome the problems associated with Wiener and Kalman filters in white Gaussian noise removal. High computational requirement is the major drawback of adaptive algorithms due to their feed forward and feedback paths. However, the integrated circuit technological advancements can be exploited in fabricating complex chips with the aid of rapidly developing very large scale integration (VLSI) technology. VLSI technological development has created a major impact in biomedical signal processing applications and filtering related applications owing to its high speed of operation and low power consumption [8]. Since the flexibility of field programmable gate arrays (FPGAs) are handy in most of the high computational signal processing applications, large number of configurable logic blocks (CLBs) are included in modern FPGAs to carry out wide range of filtering tasks [9]. In view of the fact that most of the digital signals processing (DSP) designs are targeted towards specific applications, there is a necessity to develop architectures for FPGA implementation [10]. Adaptive filtering algorithms are implemented on a single FPGA device due to rapid growth in integrated circuit fabrication. Though the single chip filters consume less power and occupy small area, the direct form architectures are slow due to the presence of error feedback path in the adaptive filter implementation. To improve the speed of filter architectures, adaptive filters are realized using pipelining and parallelism [11]. However, the filtering performance and stability of adaptive filter is severely degraded due to the incorporation of pipelining registers. Though pipelining is useful for critical path reduction, it may result in increased latency. In the VLSI implementation, the adaptive filter in its original form has a long update path, and hence the performance is relatively slow. To improve the throughput of adaptive filter, many research works have been proposed using pipelining concepts [12]. The optimal number of pipeline stages can be computed as follows:

Consider an adaptive finite impulse response (FIR) filter, which comprises of multipliers and adder trees in filtering section, error computation and coefficient update section. For a $(b \times b)$ multiplier in filtering section, $\log_2(b)$ pipeline stages are needed. An additional $\log_2(b)$ stages are required to perform coefficient multiplication. For the adder tree with level l , the required numbers of pipeline stages are $\log_2(l)$. One pipeline stage is required for the computation of the error. Hence the total numbers of pipeline stages required to achieve a maximum throughput is calculated by

$$D_{opt} = 2 \log_2(b) + \log_2(l) + 1 \quad (1)$$

Pipelining an adaptive FIR filter is relatively difficult than simple FIR filter due to the presence of feedback loop.

The stability of the adaptive filter should not be affected while pipelining. Hence the techniques used to pipeline infinite impulse response (IIR) filters have been utilized in the pipelining of adaptive FIR filter [13].

1.1 Related work

Pipelining of filter is performed using delayed model, Look-ahead transformation model, Transposed form model and Block transformation model. Mullins and Heneghan [14] presented LMS based adaptive filter architecture with delayed model for FPGA implementation. They used both the direct and transposed forms with delayed architecture implementation to obtain higher speed without a major increase in power or area [14].

The pipelined architecture of DLMS filter can be developed by incorporating pipelining registers in the forward path of FIR filter and the error feedback path in the adaptive loop. Though the pipelining registers or delay elements can reduce the critical path, they will increase the hardware complexity. Convergence and tracking performance of adaptive filter will also be affected due to pipelining [9]. Hence, it is necessary to improve the pipelined architecture by minimizing output latency and hardware complexity without affecting the filtering performance.

In the standard normalized least mean square (NLMS) algorithm, a step size parameter can be used to control the convergence performance of the filter [15]. A proper step size improves the convergence performance while reducing the MSE. Variable step size has been introduced in update equation of recursive inverse adaptive filtering for improved performance [16]. Quadratic adaptive Volterra filters have been proposed based on NLMS algorithm for increasing the speed of processing and to minimize power consumption. These filters achieve a superior convergence performance than the pipelined LMS algorithms for colored input condition. To improve the convergence with low steady state error, error normalized LMS algorithm and variable step size algorithm have been introduced [17–19]. The instantaneous data vector is normalized in NLMS algorithm, however the error vector is normalized in error normalized LMS (ENLMS).

Reconfigurable architecture for signal processing and image processing algorithms have gained huge attention among researchers to improve the performance and to minimize the power consumption [20]. Though the programmable digital signal processors are conventionally utilized for many advanced processing related applications, filtering speed and architecture can be implemented using FPGA. Hardware description languages are being utilized in FPGAs to achieve proper balance between performance and accuracy in critical biomedical signal processing applications. Hence high performance FPGA is used in this work to implement ECG signal filtering

1.2 Issues

The recursive least squares (RLS) algorithm based adaptive filtering can reduce the mean square error (MSE) at the cost of high computational complexity. Hence, it is preferred to have a white Gaussian noise removal by LMS based adaptive filtering to reduce power consumption. However, in LMS based adaptive filtering, it is necessary to have a large step size for better filtering performance.

One of the major drawbacks of the adaptive LMS algorithm is have a fixed step size which affects the filtering performance significantly. The fixed step size limitation affects filtering performance in high frequency noise removal process and white Gaussian noise removal schemes. To improve the white Gaussian noise removal performance of fixed step size LMS adaptive filters, the NLMS and ENLMS have been introduced.

1.3 Proposed method

In the proposed DENLMS algorithm, the pipelined architecture is incorporated by inserting delay elements into the feedback path so that the critical path computation time is reduced. The reduction in critical path increases the throughput of the filter without affecting the filtering performance. The DENLMS algorithm is implemented with pipelining latched using FPGA architecture. The proposed DENLMS algorithm is described using block diagram and equations in Sect. 2. Section 3 explains the VLSI architecture of DENLMS algorithm. The HDL simulation and synthesis results of ENLMS filter and DENLMS filter are presented in Sect. 4. Then, the performance of the DENLMS filter is compared with ENLMS and DLMS filters in terms of critical path, logic cells and power consumption. The conclusion of this work is provided in Sect. 5.

2 Delayed error normalized LMS algorithm

The critical path is generally reduced in VLSI implementation by pipelined structures with delay elements, so that the desired sample period can be attained [21]. In this paper, a delayed error normalized LMS (DENLMS) adaptive filter is proposed for high speed and low latency design with less computational complexity. Three main stages of delayed LMS (DLMS) algorithm are weight update, error calculation and output estimation that are given in Eqs. (2), (3) and (4) respectively [17].

$$w(n+1) = w(n) + \mu e(n-kD)x(n-kD) \quad (2)$$

where $w(n+1)$ is the updated weight, $w(n)$ is the old weight and μ is the step size. Error $e(n-kD)$ is calculated by

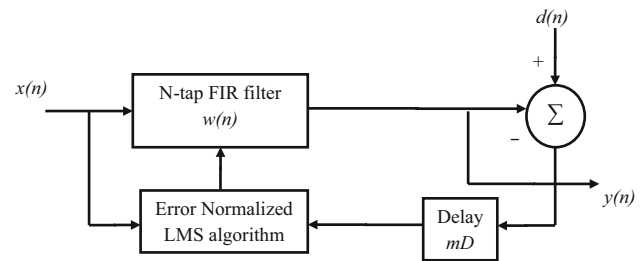


Fig. 1 Elements of proposed DENLMS algorithm

$$e(n-kD) = d(n-kD) - y(n-kD) \quad (3)$$

The output equation $y(n-kD)$ is written by the multiplication of transposed weight update coefficients and input.

$$y(n-kD) = w^T(n-kD)x(n-kD) \quad (4)$$

where kD denotes the number of delay elements utilized for pipelining stage.

$$0 < \mu < \frac{2}{(N-2k-2)(P_{TI})} \quad (5)$$

Figure 1 depicts the components of the proposed DENLMS algorithm. The step size parameter decides the filter convergence rate and the steady-state behavior of the DENLMS filter. The convergence of delayed LMS algorithm is described in the Eq. (5), using filter order N , tap-input power (P_{TI}) and pipelining stages k [9].

Based on the principles of NLMS algorithm, the weight update equation is written as

$$w(n+1) = w(n) + \mu_n e(n)x(n) \quad (6)$$

where μ_n is the normalized step size which is written in terms of [18]

$$\mu_n = \left[\frac{\mu}{p + x^T(n)x(n)} \right] \quad (7)$$

where parameter p is included to evade the small denominator value and large step size. To derive the weight update equation of ENLMS algorithm, we have to minimize the cost function

$$\|\delta w(n+1)\| = \sum_{i=0}^{N-1} (w_i(n+1) - w_i(n))^2 \quad (8)$$

under the constraint

$$w^T(n+1)x(n) = d(n) \quad (9)$$

where $\|\delta w(n+1)\|$ is the least Euclidean norm of the difference between the filter weights.

The solution of Eq. (8) is obtained by the Lagrange's multipliers method

$$J(w(n+1), \lambda) = \|\delta w(n+1)\|^2 + \lambda(d(n) - w^T(n+1)x(n)) \quad (10)$$

The minimum of $J(w(n+1), \lambda)$ is obtained by setting

$$\frac{\partial J(w(n+1), \lambda)}{\partial w_j(n+1)} = 0 \quad (11)$$

Using Eq. (11), we obtain

$$w_j(n+1) = w_j(n) + \frac{1}{2}\lambda x(n-j) \quad (12)$$

and

$$\lambda = \frac{2e(n)}{\sum_{i=0}^{N-1} (e(n-i))^2} \quad (13)$$

To introduce the extra degree of freedom, the parameter μ is introduced

$$w_j(n+1) = w_j(n) + \frac{\mu}{p + \|e(n)\|^2} e(n)x(n-j) \quad (14)$$

Equation (14) is written in another form as

$$w(n+1) = w(n) + \frac{\mu}{p + e^t(n)e(n)} e(n)x(n) \quad (15)$$

Hence, we have

$$w(n+1) = w(n) + \mu_{en} e(n)x(n) \quad (16)$$

The normalized step size of ENLMS algorithm is written based on squared norm of error vector instead of input vector [7]. The error normalized step size is written as

$$\mu_{en} = \left[\frac{\mu}{p + e^t(n)e(n)} \right] \quad (17)$$

where parameter p is set to avoid the high denominator value and small step size.

From Eq. (17), if the error is high, the value of step size decreases which may affect the convergence. Conversely, the increase in step size parameter leads to faster convergence, since $e(n)$ decreases in magnitude. The algorithm initially converges slowly due to small μ value. In Eq. (17), variable step size depends only on the error $e(n)$. The necessary condition for filter to be stable is μ_{en} should lie in the range $(0, \mu)$. When $e^2(n)$ is equal to zero, $\mu_{en} = \mu$. But the high $e^2(n)$ values decrease the step size and affect the convergence. To avoid that, it is necessary to choose the negative p value. Hence the stability condition is $p < 0$. Based on

the stability condition of NLMS algorithm [15], the lower on the parameter p is determined by considering the uniform convergence condition

$$e(n+1) \leq |1 - \mu_{en}| |e(n)|^2 \quad (18)$$

Hence

$$|1 - \mu_{en}| |e(n)|^2 < 1 \quad (19)$$

which gives

$$0 < \mu / |e(n)|^2 + p < 2 / |e(n)|^2 \quad (20)$$

From Eq. (20), the lower bound of stability is given by

$$p > -\frac{|e(n)|^2}{2} \quad (21)$$

In the ENLMS algorithm, the weight update equation is written similar to NLMS algorithm as

$$w(n+1) = w(n) + \mu_{en} e(n)x(n) \quad (22)$$

The normalized step size of ENLMS algorithm is written based on squared norm of error vector instead of input vector [20]. The error normalized step size is written as

$$\mu_{en} = \left[\frac{\mu}{p + e^t(n)e(n)} \right] \quad (23)$$

where parameter p is incorporated to lessen the high denominator value and small step size.

Table 1 compares the computational complexity of DENLMS algorithm with NLMS algorithm, transform domain least mean square (TDLMS) algorithm and DNLMS algorithms. From the Table 1, it is observed that TDLMS algorithm needs $6N + 1$ multiplications and $3N$ additions whereas NLMS algorithm utilizes only $3N$ multiplications and $3N$ additions. Hence the multiplication operations in NLMS algorithm are almost reduced to half of the TDLMS algorithm. NLMS, DNLMS and DENLMS algorithms require $3N$ multiplications and $3N + 1$ additions. Hence, NLMS, DNLMS and DENLMS require less number of additions and multiplications while comparing with TDLMS algorithm. The DNLMS algorithm and proposed

Table 1 Comparison of computations

Algorithm	NLMS	TDLMS	DNLMS	DENLMS
Multiplications	3N	6N+1	3N	3N
Additions	3N+1	3N	3N+1	3N+1
Delay	0	0	1	1

DENLMS need extra delay element for pipelining purpose which makes the algorithm faster and useful in low power implementation.

3 VLSI architecture for DENLMS algorithm

The LMS algorithms are comparatively simpler than other adaptive algorithms and the total number of required computations is $2N$ additions and $2N + 1$ multiplications. The simple LMS weight update equation is written as

$$w(n) = w(n-1) + \mu e(n)u(n) \quad (24)$$

VLSI architectures are developed using pipelining and parallel processing concepts. Pipelined interleaving and look-ahead decomposition are the two important methodologies used for signal processing applications and filters [22,23]. Parhi [11] introduced pipelined interleaving and parallelism for recursive digital filter applications [11]. Two techniques used are scattered look-ahead decomposition and block filtering method. Applying the look-ahead decomposition in Eq. (24) leads to

$$w(n) = w(n-m_2) + \sum_{i=0}^{m_2-1} \mu e(n-i)u(n-i) \quad (25)$$

where m_2 denotes the number of delay elements used in the weight update loop.

From Eq. (11), it can be noted that there are m_2 latches in the weight update loop due to transformation. However, look-ahead transformation can be done exactly by expressing $e(n)$ as a function of $w(n-m_2)$. The transformed equation is substituted in Eq. (11), which leads to complicated equation and increases hardware complexity. Though the equation for look ahead transformation is complex, it is possible to reduce the computational complexity by introducing relaxed look-ahead principle. In this paper delayed relaxation is applied which leads to m_1 delay elements in the error feedback path of adaptive filter [9]. Applying the relaxed look-ahead, we get

$$w(n) = w(n-m_2) + \mu \sum_{n=0}^{m_2-1} e(n-m_1-i)u(n-m_1-i) \quad (26)$$

For the above summation, the important assumption is that the gradient estimate $e(n).u(n)$ does not vary much over m_1 samples. From Eq. (12), it can be noted that the hardware overhead due to transformation is $n(m_2-1)$ adders and it can be further reduced using sum relaxation by taking m_2^1 terms, where $m_2^1 \leq m_2$

$$w(n) = w(n-m_2) + \sum_{i=0}^{m_2^1} \mu e(n-m_1-i)u(n-m_1-i) \quad (27)$$

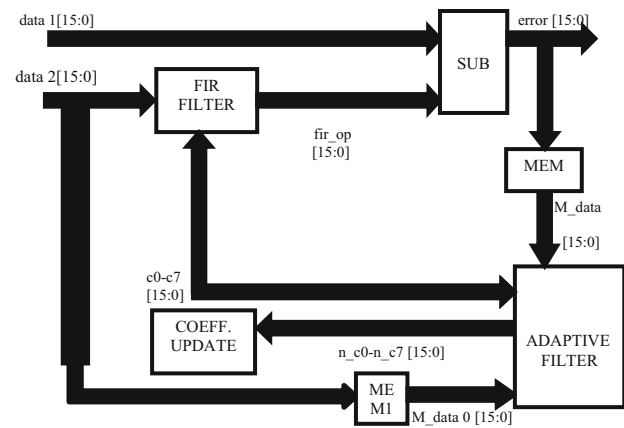


Fig. 2 Architecture of DENLMS adaptive filter

This relaxation is valid for a slowly varying product $e(n)u(n)$. Therefore, $e(n)$ can be written as

$$e(n) = d(n) - [w^T(n-m_2-1) + \mu \sum_{i=0}^{m_2^1-1} e(n-m_1-i-1)u(n-m_1-i-1)]u(n) \quad (28)$$

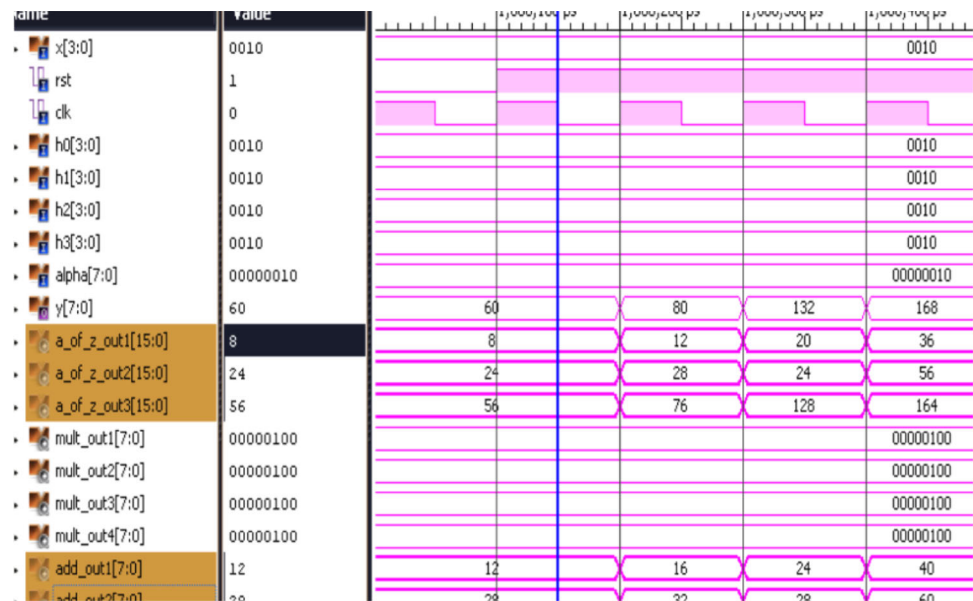
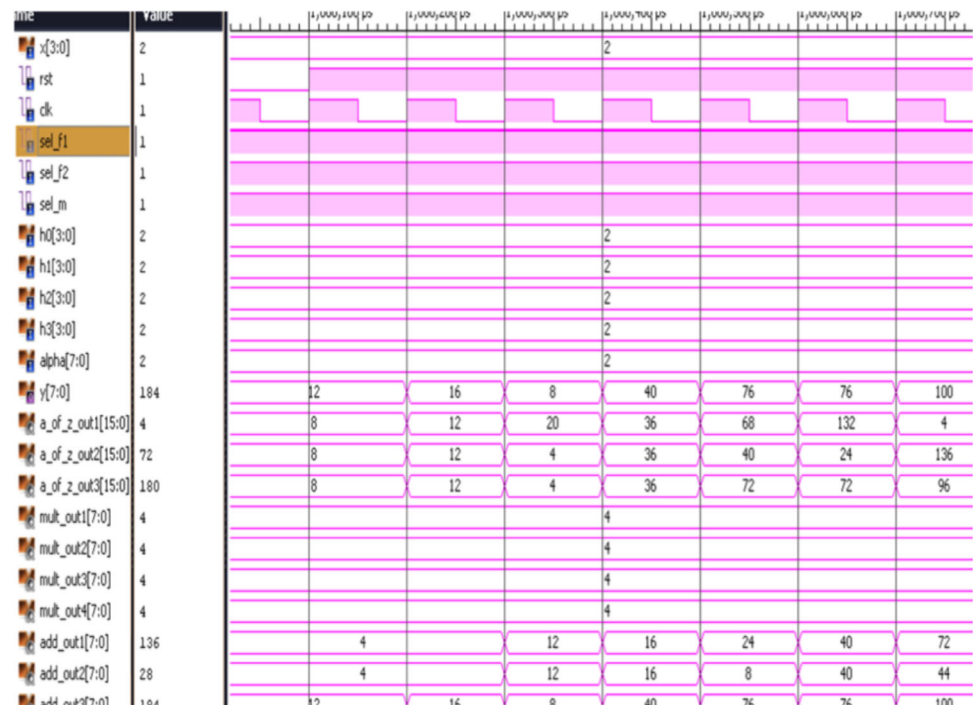
Assuming μ is sufficiently small and replacing $w(n-m_2-1)$ by $w(n-m_2)$

$$e(n) = d(n) - w^T(n-m_2)u(n) \quad (29)$$

The above equation is applied to the proposed DENLMS adaptive filter. Figure 2 describes the pipelined DENLMS architecture with 16-bit data. The ‘data1’ is the desired output and ‘data2’ is the input applied to the FIR filter. Error signal is produced by the ‘SUB’ block based on the ‘data1’ and filter output ‘fir_op’. The adaptive filtering block generates the updated coefficient which is applied to FIR filter to reduce the error. The adaptive filter in the block diagram utilizes the DENLMS algorithm.

4 Results and discussion

The pipelined architectures generally improve the filter performance and minimize the power consumption of the system. In the proposed work, DENLMS algorithm with pipelined architecture is implemented in FPGA. The error reducing or noise removing capability of any system is checked by VLSI simulation process and the designed system can be viewed by the synthesis process. Simulation, synthesis and implementation results are obtained for both DENLMS and ENLMS filters for comparison purpose. Verilog hardware description language is utilized in this work

Fig. 3 Simulation results of ENLMS architecture**Fig. 4** Simulation results of pipelined DENLMS architecture

due to large number of libraries and integrated circuit related functions.

Modelsim tool is used for simulation experiments and Xilinx Integrated Software Environment (ISE) is utilized for synthesis and implementation by considering Virtex 5 as target chip. Xilinx synthesis tool (XST) converts the design code into synthesizable register transfer level (RTL) diagram with gate level net list. Once the design is synthesized, the design is ready to be implemented in FPGA chip. Synthesized design is partitioned and floor planned in Xilinx ISE environment. The floor planned design is placed and routed manually to

obtain optimized implementation. Xilinx ISE provides the area report, incurred delay in the implementation with power report.

In DENLMS and ENLMS based adaptive filters, the filter coefficients or weights of the adaptive FIR filter are updated by applying DENLMS algorithm and ENLMS algorithm respectively. Figures 3 and 4 show Modelsim simulation results obtained after eighth iteration for ENLMS filter and DENLMS filter respectively. Verilog HDL programming is done in the data flow modeling with input, reset, clock, error and output signals. After resetting the circuit, the filter gener-

Table 2 Comparison of device utilization

Logic utilization	ENLMS	DNLMS	DENLMS
Number of slice flip flops	5213	6313	6328
Number of 4 input LUTs	6998	7232	7243
Number of occupied cells	3867	4112	4215
Total number of 4 input LUTs	7213	8725	8790
Number of bonded IOBs	120	120	120
Number of BUFGMUXs	4	4	4

ates the error output according to the input and clock signals. The obtained error for ENLMS filter at eighth iteration is 38,403 and for DENLMS filter is 34,647 with pipelined architecture with the assumption of 16 bit value. By comparing DENLMS and ENLMS filters, pipelined DENLMS algorithm reduces error value significantly. Since the error value is much higher in DNLMS filter while comparing to DENLMS and ENLMS filters, the DNLMS simulation result is not included.

The next step is to convert the RTL code to synthesizable design with gate level net list, targeted to a specific Virtex 5 FPGA XC5LVX330 chip. This step in design flow is known as synthesis. Optimization constraints comprise of input and output delays and operating frequency (clock frequency). Once the design is synthesized, the next step is to implement the design after timing verification. Implementation of the design involves floor planning, placement and routing. RTL schematic diagram of ENLMS filter and DENLMS filter are obtained in synthesis process. The synthesis process converts the data flow level Verilog HDL into RTL level gates and gate net list. The synthesized DENLMS filter includes extra delay element in the error feedback architecture. After completing synthesis process, device utilization summary can be obtained. A comparison is made between the proposed algorithm based filter and other filters. Table 2 compares the device utilization summary of different filter implementations.

Based on the logic utilization, it can be concluded that both the DNLMS and DENLMS require more number of flip flops and look up table while comparing with ENLMS algorithm. Results indicate that more number of logic elements are required in DNLMS and DENLMS due to the insertion of extra delay in the critical path. Synthesized design of ENLMS, DNLMS and DENLMS filters are implemented in FPGA to analyze the area, delay and power requirements.

The performance comparison is done using DENLMS algorithm, ENLMS algorithm and DNLMS algorithms. Slack, maximum critical paths, number of ports, number of nets, total logic cell, total dynamic power and cell leakage power are calculated for comparison purpose. Slack is one of the important timing parameter which is based on the arrival

Table 3 Comparison of ENLMS, DNLMS and DENLMS filters

Parameters	ENLMS	DNLMS	DENLMS
Maximum critical paths	56	304	322
Number of ports	40	40	40
Number of nets	110	120	160
Total logic cells	30465	32859	33546
Total dynamic power (mW)	74	64	42
Cell leakage power (μ W)	154	132	107

time difference between the actual time and the desired time for a timing path. For a specific timing path, it is useful to identify whether the design is operated at the specific speed. Since the slack value is positive for all the designed filters, the timing requirements are met. The number of critical path is high in DNLMS and DENLMS filters due to the presence of pipelining latches.

The implementation of DNLMS architecture has 40 I/O ports, 120 nets and 328,594 logic cells. However the nets, logic cells and ports are comparatively less in ENLMS algorithm due to the absence of delay elements or pipelining latches. The pipelined DENLMS filter has 40 I/O ports, 160 nets and 335,468 logic cells. The increase in the number of nets and logic cells is due to the added memory elements in pipelined architecture. The total area is increased by 20.4% in DENLMS filter while comparing with ENLMS filter. The increase in area of the design is due to the high number of logic cells. However, the filter performance is improved due to slack reduction. The utilized ports for all the three filter DNLMS, ENLMS and DENLMS are same. The obtained slack, total dynamic power and leakage power are listed in Table 3 by implementing these filters. The Slack value of ENLMS, DNLMS and DENLMS algorithm are 0.27, 0.31 and 0.19 respectively. The slack value reduction of 35.24% is achieved in DENLMS filter. Total dynamic power of DENLMS filter is 42 mW which is less than 74 mW of ENLMS filter and 64 mW of DNLMS filter. The cell leakage power of DENLMS filter is 107 μ W which is less than 154 μ W of ENLMS filter and 132 μ W of DNLMS filter.

5 Conclusion

In this work, the performance improvement and power reduction are achieved using DENLMS algorithm. The computational complexity of DENLMS is slightly higher than conventional DNLMS and ENLMS algorithms due to the inclusion extra delay element for the purpose of pipelining. The simulation, synthesis and implementation results are obtained for VLSI implementation of ENLMS filter,

DNLMS filter and DENLMS filters using Virtex 5 FPGA XC5LVX330 chip. The nets, logic cells and power consumption are less in ENLMS adaptive filter due to the absence of delay element. The pipelined DNLMS and DENLMS filters require more area compared to the pipeline less ENLMS filter. However, the pipelined architecture of DENLMS filter increases the performance while reducing the power consumption. The total area is increased by 20.4% in DENLMS filter, but the achieved total dynamic power reduction and cell leakage power reduction are 31.8 and 33.5% respectively. From the obtained results, pipelined filters increase the area of the design due to the presence of latches. However, significant amount of power savings are achieved using pipelining.

References

- Hung, K., Zhang, Y.T.: Implementation of a WAP-based telemedicine system for patient monitoring. *IEEE Trans. Inf. Technol. Biomed.* **7**, 101–107 (2003)
- Leski, J.M., Henzel, N.: ECG baseline wander and powerline interference reduction using nonlinear filter bank. *Signal Process.* **35**, 781–793 (2004)
- Elhossini, A., Areibi, S., Dony, R.: An FPGA implementation of the LMS adaptive filter for audio processing. In: *IEEE International Conference on Reconfigurable Computing and FPGAs*, pp. 1–8 (2006)
- El-Dahshan, E.S.A.: Genetic algorithm and wavelet hybrid scheme for ECG signal denoising. *Telecommun. Syst.* **46**, 209–215 (2011)
- Chang, K.M., Liu, S.H.: Gaussian Noise filtering from ECG by Wiener filter and ensemble empirical mode decomposition. *J. Signal Process. Syst.* **64**, 249–264 (2011)
- Vaseghi, S.V.: *Advanced digital signal processing and noise reduction*, 3rd edn. Wiley, England (2006)
- Sayadi, O., Shamsollahi, M.B.: ECG denoising and compression using a modified extended Kalman filter structure. *IEEE Trans. Biomed. Eng.* **55**, 2240–2248 (2008)
- Weste, N., Harris, D.: *CMOS VLSI design*. Addison Wesley, Boston (2004)
- Ting, L., Woods, R., Cowan, C.: Virtex FPGA implementation of a pipelined adaptive LMS predictor for electronic support measures receivers. *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.* **13**, 86–95 (2005)
- Dempster, A., Macleod, M.: Use of minimum-adder multiplier blocks in FIR digital filters. *IEEE Trans. Circuits Syst. II* **42**, 569–577 (1995)
- Parhi, K.K.: *VLSI digital signal processing systems: design and implementation*. Wiley, New York (2007)
- Long, G., Ling, F., Proakis, J.: The LMS Algorithm with delayed coefficient adaptation. *IEEE Trans. Acoust. Speech Signal Process.* **37**, 1397–1405 (1989)
- Meyer-Baese, U.: *Digital signal processing with field programmable gate arrays*. Springer, Heidelberg (2007)
- Mullins, S., Heneghan, C.: Alternative least mean square adaptive filter architectures for implementation on field programmable gate arrays. In: *11th IEEE European Conference in Signal Processing*, pp. 1–4 (2002)
- Gupta, A., Joshi, S.: Variable step-size LMS algorithm for fractal signals. *IEEE Trans. Signal Process.* **56**, 1411–1420 (2008)
- Satheeskumaran, S., Sabirigiriraj, M.: VLSI implementation of a new LMS based algorithm for noise removal in ECG signal. *Int. J. Electron.* **103**, 975–984 (2016)
- Ahmad, M., Kukrer, O., Hoccanin, A.: Recursive inverse adaptive filtering algorithm. *Digit. Signal Process.* **21**, 491–496 (2011)
- Satheeskumaran, S., Sabirigiriraj, M.: A new LMS based noise removal and DWT based R-peak detection in ECG signal for biotelemetry applications. *Natl Acad. Sci. Lett.* **37**(4), 341–349 (2014)
- Kohler, B.U., Hennig, C., Orglmeister, R.: The principles of software QRS detection. *IEEE Eng. Med. Biol. Mag.* **21**, 42–57 (2002)
- Meena, S.M., Linganagouda, K.: Implementation and analysis of optimized architectures for rank order filter. *J. Real-Time Image Process.* **3**, 33–41 (2008)
- Rahman, M.Z.U., Shaik, R.A., Reddy, D.V.: Efficient and simplified adaptive noise cancellers for ECG sensor based remote health monitoring. *IEEE Sens. J.* **12**, 566–573 (2012)
- Van, L.D., Chang, C.H.: Pipelined RLS adaptive architecture using relaxed givens rotations (RGR). In: *IEEE International Symposium on Circuits and Systems*, pp. 1–3 (2002)
- Jubairahmed, L., Satheeskumaran, S., Venkatesan, C.: Contourlet transform based adaptive nonlinear diffusion filtering for speckle noise removal in ultrasound images. *Clust. Comput.* 1–10 (2017). <https://doi.org/10.1007/s10586-017-1370-x>



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